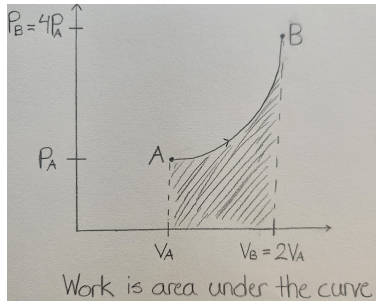


Problem 1 Solution

a)

$$P = CV^2 \text{ with } C = \frac{P_A}{V_A^2}$$

$$W = \int P dV = \int_{V_A}^{2V_A} CV^2 dV = \frac{7}{3} CV_A^3 = \frac{7}{3} P_A V_A$$



b)

$$\Delta U = nC_v \Delta T = 3nR \Delta T \quad \text{for polyatomic gas}$$

$$PV = nRT \Rightarrow T = \frac{PV}{nR}$$

$$\Delta U = 3nR \left(\frac{P_B V_B}{nR} - \frac{P_A V_A}{nR} \right)$$

$$= 3(P_B V_B - P_A V_A)$$

$$\frac{P_A}{V_A^2} = \frac{P_B}{V_B^2} \Rightarrow P_B = 4P_A$$

$$\Delta U = 3(4P_A \cdot 2V_A - P_A V_A) = 21P_A V_A$$

c)

$$\Delta U = Q - W$$

$$Q = \Delta U + W = \frac{70}{3} P_A V_A$$

d)

$$\Delta U = -W \quad \text{neglecting } Q$$

$$\Delta U = 3(P_C V_C - P_B V_B)$$

$$= 3(5P_A \cdot 2V_A - 4P_A \cdot 2V_A)$$

$$= 6P_A V_A$$

$$W = -6P_A V_A$$

We took the gas out of thermal equilibrium with the environment and $W = \int P dV$ only holds for reversible processes. We should therefore expect to do some nonzero amount of work on the gas.

e)

Consider the reversible path between states B and C , where $W = 0$.

$$\Delta U = Q$$

$$dQ = dU = nC_v dT = 3nR dT$$

$$\begin{aligned}\Delta S &= \int \frac{dQ}{T} \\ &= 3nR \int_{\frac{8P_A V_A}{nR}}^{\frac{10P_A V_A}{nR}} \frac{dT}{T} \\ &= 3nR \ln \frac{5}{4}\end{aligned}$$

Above is the entropy change of the gas. The entropy change of the environment is 0 because the process was too sudden for heat transfer to take place, so the total entropy change is the same as above. The positive sign confirms irreversibility.

Problem 1 Rubric

a.

- +1 pt: Valid PV-diagram drawn
- +1 pt: Correct area identified
- +1 pt: $W = \int P dV$
- +1 pt: $W = \frac{7}{3}P_A V_A$

b.

- +1 pt: $E_{int} = nC_V T$
- +1 pt: $C_V = 3R$
- +1 pt: $PV = nRT$
- +1 pt: $P_B = 4P_A$
- +1 pt: $\Delta E_{int} = 21P_A V_A$
- +1 pt: Fully correct/partial credit for $\Delta E_{int} = Q - W$

c.

- +1 pt: $\Delta E_{int} = Q - W$
- +1 pt: $Q = \frac{70}{3}P_A V_A$ /correct Q based on $a.$ and $b.$

d.

- +1 pt: $\Delta E_{int} = Q - W$
- +1 pt: $E_{int} = nC_V T$
- +1 pt: $PV = nRT$
- +1 pt: $Q = 0$
- +1 pt: $W = -6P_A V_A$
- +1 pt: Mentions irreversibility allows $W \neq 0$

e.

- +1 pt: $\Delta S_{gas} = \int \frac{dQ}{T}$
- +1 pt: $\Delta E_{int} = Q - W$
- +1 pt: $W = 0$ for reversible path between same endpoints
- +1 pt: $E_{int} = nC_V T$
- +1 pt: $\Delta S_{gas} = 3nR \ln \frac{5}{4}$
- +1 pt: $\Delta S_{env} = 0$
- +1 pt: Mentions positive sign shows irreversibility

Problem 3 Solution

(a)

The top semicircle has resistance R , while the bottom semicircle has resistance $2R$. Since the two branches are in parallel, the current divides inversely proportional to resistance:

$$I_1/I_2 = \frac{1}{R}/\frac{1}{2R} = 2$$

Using $I_1 + I_2 = I$,

$$\boxed{I_1 = \frac{2I}{3}, \quad I_2 = \frac{I}{3}}$$

(b)

Using Biot-Savart law, the magnetic field at the center from a semicircle of radius a carrying current I is

$$B = \frac{\mu_0 I}{4a}$$

The top current is clockwise, so its field points into the page.

The bottom current is counterclockwise, so its field points out of the page.

Thus,

$$B_{\text{net}} = \frac{\mu_0}{4a}(I_1 - I_2)$$

Substituting the currents,

$$B_{\text{net}} = \frac{\mu_0}{4a} \left(\frac{2I}{3} - \frac{I}{3} \right) = \frac{\mu_0 I}{12a}$$

$$\boxed{B_{\text{net}} = \frac{\mu_0 I}{12a}}$$

Direction: into the page.

The straight wires contribute zero magnetic field at the center because the center lies along their line.

(c)

The two semicircular resistors are in parallel:

$$R_{\text{eq}} = 1/(1/R + 1/(2R)) = \frac{R(2R)}{R + 2R}$$

$$\boxed{R_{\text{eq}} = \frac{2R}{3}}$$

(d)

The two capacitors are in parallel:

$$C_{\text{eq}} = C + 2C = 3C$$

The time constant is

$$\tau = R_{\text{eq}}C_{\text{eq}}$$

Substituting,

$$\tau = \left(\frac{2R}{3}\right)(3C) = 2RC$$

$$\boxed{\tau = 2RC}$$

(e)

For a charging RC circuit,

$$V_R = \mathcal{E}e^{-t/\tau}$$

and

$$V_C = \mathcal{E}(1 - e^{-t/\tau})$$

Set them equal:

$$\mathcal{E}e^{-t/\tau} = \mathcal{E}(1 - e^{-t/\tau})$$

Cancel $\mathcal{E}$:

$$e^{-t/\tau} = 1 - e^{-t/\tau}$$

$$2e^{-t/\tau} = 1$$

$$e^{-t/\tau} = \frac{1}{2}$$

Taking the logarithm,

$$-\frac{t}{\tau} = \ln\left(\frac{1}{2}\right)$$

$$t = \tau \ln 2$$

$$\boxed{t = \tau \ln 2}$$

Grading Rubric (25 pts)

(a) 4 pts

- 1 pt: Recognizes the two semicircles are in parallel
- 2 pt: States that $R_2 = 2R_1$ due to resistivity (or equivalent)
- 1 pt: Uses current division/inverse-resistance relation correctly

(b) 7 pts

- 2 pts: Uses Biot-Savart law correctly
- 2 pt: Recognizes straight wires give zero contribution at center
- 2 pts: Correct directions from right-hand rule (top: into page bottom: out of page)
- 1 pt: Correct final magnitude

(c) 4 pts

- 2 pts: Correct formula for total resistance of parallel resistors
- 2 pts: Correct final answer R_{eq}

(d) 4 pts

- 2 pts: Correct capacitor combination:

$$C_{eq} = 3C$$

- 2 pts: Final answer:

$$\tau = 2RC$$

(e) 6 pts

- 2 pt: Recognizes the circuit behaves like an equivalent RC circuit
- 2 pt: States that V_R is proportional to $e^{-t/\tau}$ (or equivalent)
- 2 pt: Correct final answer:

$$t = \tau \ln 2$$

Total: 25 pts

Problem 4

a. **Chosen Method:** Ampère's Law. Due to the cylindrical symmetry and the assumption $b \gg a$, we can treat the field as that of an ideal solenoid.

- **Direction:** According to the Right-Hand Rule, the magnetic field $\vec{B}$ is directed along the symmetry axis, specifically in the $+\hat{z}$ direction.
- **Magnitude:** Outside the cylinder ($r > a$), the field is zero ($B = 0$). Inside the cylinder ($r < a$), we use a rectangular Amperian loop of length ℓ .

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{encl} \implies B \cdot \ell = \mu_0 (K \cdot \ell) \implies B = \mu_0 K \quad (1)$$

Final Result:

$$\vec{B}(r) = \begin{cases} \mu_0 K \hat{z} & \text{for } r < a \\ \vec{0} & \text{for } r > a \end{cases} \quad (2)$$

b. The surface current density varies as: $K(t) = K_0 \left(1 + \frac{t}{T}\right)$. Consequently, the magnetic flux varies over time. According to Faraday's Law: $\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi}{dt}$.

Inside the cylinder ($r < a$):

The flux through a loop of radius r is $\Phi(t) = B(t) \cdot \pi r^2 = \mu_0 K_0 \left(1 + \frac{t}{T}\right) \pi r^2$.

$$E(2\pi r) = -\frac{d}{dt} \left[\mu_0 K_0 \left(1 + \frac{t}{T}\right) \pi r^2 \right] = -\frac{\mu_0 K_0}{T} \pi r^2 \quad (3)$$

$$\boxed{E = -\frac{\mu_0 K_0 r}{2T} \quad \text{for } r < a} \quad (4)$$

Outside the cylinder ($r > a$):

The magnetic field is zero for $r > a$, but the loop still encloses the flux inside the cylinder. The total flux is $\Phi(t) = \mu_0 K_0 \left(1 + \frac{t}{T}\right) \pi a^2$.

$$E(2\pi r) = -\frac{\mu_0 K_0}{T} \pi a^2 \implies \boxed{E = -\frac{\mu_0 K_0 a^2}{2Tr} \quad \text{for } r > a} \quad (5)$$

Direction:

According to Lenz's law, the induced field must oppose the increase of the magnetic flux. Therefore the electric field is in the direction opposed to the one of K (clockwise when $\hat{z}$ points out of the page)

c. According to Lenz's Law, the induced effects must oppose the increase in magnetic flux.

- The primary current K is increasing, so $\vec{B}$ is increasing in the $+\hat{z}$ direction.
- The induced magnetic field must point in the $-\hat{z}$ direction to oppose this change.
- Therefore, the induced current must flow in the direction opposite to K , which is the $-\hat{\phi}$ direction.

Since $\vec{j} = \sigma \vec{E}$, the induced current follows the same direction as the induced electric field $\vec{E}$.

d. **Sign Convention:** We choose the path $d\vec{l}$ in the direction of K ($+\hat{\phi}$), which implies the area vector $d\vec{A}$ points in the $+\hat{z}$ direction.

$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -\frac{\mu_0 \pi a^2 K_0}{T} \quad (6)$$

The negative sign confirms that the induced EMF (and current) points opposite to $d\vec{l}$. The absolute value of the induced current in the tube with resistance R is:

$$I_{ind} = \frac{|\mathcal{E}|}{R} = \frac{\mu_0 \pi a^2 K_0}{RT} \quad (7)$$

- e. By definition, $L = \frac{\Phi_{total}}{I}$, where I is the total current passing through the inductor. For a tube of length b , the total current circulating around the circumference is $I = K \cdot b$. The total magnetic flux through the cross-section $S = \pi a^2$ is:

$$\Phi = B \cdot S = (\mu_0 K) \cdot (\pi a^2) = \mu_0 \left(\frac{I}{b} \right) \pi a^2 \quad (8)$$

Solving for the self-inductance:

$$L = \frac{\mu_0 \pi a^2}{b} \quad (9)$$

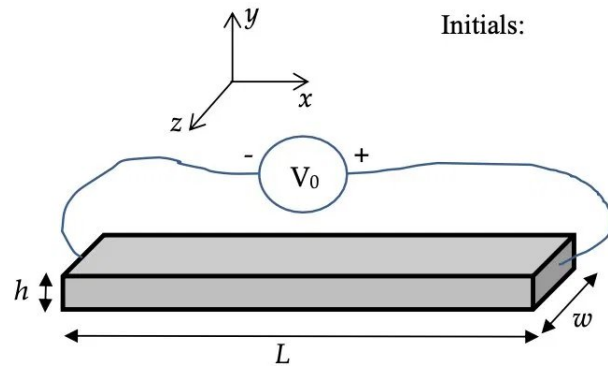
Problem 5 – Conducting Ribbon & Hall Effect

Solutions and Grading Rubric (25 points)

Problem Setup

Problem 5 – 25 points

Consider a conducting ribbon of length L , width w and height h . The material used to make the ribbon is known to have a resistivity ρ , a number of atoms per unit volume d_A , and two free electrons (individual charge $-e$) per atom. A voltage V_0 is measured experimentally across the ribbon – with higher electric potential on the right – as a uniform current is passed along the length.



Convention note (important for grading direction): Since the higher potential is on the right ($+\hat{x}$ end), conventional current flows *inside the ribbon* from high to low potential, i.e. in the $-\hat{x}$ direction. Some students may instead define $+\hat{x}$ as the current direction; **accept either convention as long as it is used consistently throughout the problem.**

Part (a) – Resistance of the Ribbon

[4 pts]

Solution

The cross-sectional area perpendicular to the current is $A = wh$, and the length along the current direction is L . Using $R = \rho L/A$:

$$R = \frac{\rho L}{wh}$$

Explanation & Reasoning

The resistance of a uniform conductor follows directly from the definition of resistivity. The cross-section perpendicular to the current flow is the $w \times h$ rectangle (since current flows along the length L). Students must correctly identify which dimension is the length (parallel

to current) and which form the cross-section.

Grading Rubric

- **+2 pts** – Correct formula $R = \rho L/A$ stated or used.
- **+1 pt** – Correctly identifies cross-sectional area as $A = wh$.
- **+1 pt** – Final answer $R = \rho L/(wh)$ with correct algebra.

Part (b) – Current Density $\vec{j}$

[4 pts]

Solution

Current through the ribbon:

$$I = \frac{V_0}{R} = \frac{V_0 wh}{\rho L}$$

Magnitude of current density:

$$j = \frac{I}{A} = \frac{I}{wh} = \frac{V_0}{\rho L}$$

Direction: Conventional current flows from high to low potential inside the ribbon, i.e. from the $+\hat{x}$ end (right) to the $-\hat{x}$ end (left):

$$\vec{j} = -\frac{V_0}{\rho L} \hat{x}$$

Equivalently, this follows directly from Ohm's law in local form: $\vec{j} = \vec{E}/\rho$, with $\vec{E}$ pointing from $+$ to $-$ inside the ribbon.

Explanation & Reasoning

Two valid paths to the answer:

1. Compute $I = V_0/R$, then $j = I/A$.
2. Use local Ohm's law: $E = V_0/L$ (uniform field), so $j = E/\rho = V_0/(\rho L)$.

The direction must reflect that inside a passive conductor, $\vec{j}$ points from high to low potential.

Grading Rubric

- **+1 pt** – Correct formula $j = I/A$ or $j = E/\rho$.
- **+2 pts** – Correct magnitude $j = V_0/(\rho L)$.
- **+1 pt** – Correct direction ($-\hat{x}$, or stated equivalently as “from $+$ to $-$ inside the ribbon”).

Part (c) – Drift Velocity $\vec{v}_d$

[5 pts]

Solution

The number density of free electrons is

$$n = 2 d_A \quad (2 \text{ electrons per atom})$$

From $\vec{j} = nq\vec{v}_d$ with $q = -e$:

$$\vec{v}_d = \frac{\vec{j}}{nq} = \frac{\vec{j}}{-2d_A e}$$

Substituting $\vec{j} = -\frac{V_0}{\rho L}\hat{x}$:

$$\vec{v}_d = \frac{V_0}{2d_A e \rho L} \hat{x}$$

Direction: The electrons drift in the $+\hat{x}$ direction, *opposite* to the conventional current (as expected for negative charge carriers).

Explanation & Reasoning

Two pieces must be combined:

1. Carrier density: “2 free electrons per atom” means $n = 2d_A$, not $n = d_A$. Missing this factor of 2 is the most common error.
2. Sign of charge: since $q = -e < 0$, electrons drift *opposite* to $\vec{j}$.

Grading Rubric

- **+1 pt** – Correctly states $n = 2d_A$.
- **+1 pt** – Uses correct relation $\vec{j} = nq\vec{v}_d$ (or equivalent).
- **+2 pts** – Correct magnitude $v_d = V_0/(2d_A e \rho L)$.
- **+1 pt** – Correct direction ($+\hat{x}$, opposite to $\vec{j}$).

Part (d) – Forces on Free Charges in $\vec{B}_0$

[6 pts]

Solution

Initial state (transient). Each free electron moving with velocity $\vec{v}_d = v_d \hat{x}$ (where $v_d = \frac{V_0}{2d_A e \rho L}$ from part (c)) in field $\vec{B}_0 = -B_0 \hat{y}$ experiences the magnetic Lorentz force:

$$\vec{F}_B = q \vec{v}_d \times \vec{B}_0 = (-e)(v_d \hat{x}) \times (-B_0 \hat{y}) = e v_d B_0 (\hat{x} \times \hat{y}) = e v_d B_0 \hat{z}$$

Substituting v_d in terms of given variables:

$$\vec{F}_B = \frac{V_0 B_0}{2 d_A \rho L} \hat{z}$$

So electrons are deflected in the $+\hat{z}$ direction.

Charge accumulation. Electrons accumulate on the $+\hat{z}$ face of the ribbon, leaving the $-\hat{z}$ face with a net positive charge (deficit of electrons).

Induced Hall electric field. The charge separation creates a transverse electric field $\vec{E}_H$ pointing from the positive face ($-\hat{z}$) to the negative face ($+\hat{z}$):

$$\vec{E}_H = E_H \hat{z}$$

This field exerts an *electric force* on each electron:

$$\vec{F}_E = (-e)\vec{E}_H = -eE_H \hat{z}$$

which opposes the magnetic force.

Steady-state outcome. Charge keeps accumulating until $\vec{F}_E + \vec{F}_B = 0$:

$$eE_H = e v_d B_0 \quad \Rightarrow \quad E_H = v_d B_0 = \frac{V_0 B_0}{2 d_A e \rho L}$$

After equilibrium, electrons drift purely along $\hat{x}$ again – the longitudinal current is unchanged, but a transverse **Hall voltage** now exists across the width.

Explanation & Reasoning

Students must identify *both* forces (magnetic and induced electric) and explain the dynamical picture: transient deflection $\rightarrow$ charge buildup $\rightarrow$ induced E -field $\rightarrow$ equilibrium. A correct answer connects all four pieces.

Cross-product sanity check: $\hat{x} \times \hat{y} = \hat{z}$, and the two minus signs (from $q = -e$ and $\vec{B} = -B_0 \hat{y}$) cancel, giving $+\hat{z}$ for the magnetic force on the electrons. This is consistent with the right-hand rule applied to conventional current ($-\hat{x}$) crossed with $\vec{B}$ ($-\hat{y}$), giving $\vec{F}$ on positive carriers in $-\hat{z}$ – and the equivalent picture for negative carriers (electrons) is $+\hat{z}$.

Grading Rubric

- **+2 pts** – Correctly identifies magnetic Lorentz force $\vec{F}_B = q\vec{v} \times \vec{B}$ with correct cross-product evaluation (direction $+\hat{z}$ on electrons).
- **+1 pt** – States that electrons accumulate on one face ($+\hat{z}$), leaving the opposite face positive.
- **+1 pt** – Identifies the induced transverse electric field (Hall field) and the corresponding electric force on the electrons.
- **+1 pt** – Explains that equilibrium is reached when the two forces balance: $eE_H = ev_d B_0$.
- **+1 pt** – States that in steady state, the longitudinal current is unchanged (electrons

resume drift along $\hat{x}$ only).

Part (e) – Hall Voltage

[6 pts]

Solution

At equilibrium, $E_H = v_d B_0$ across the width w . The Hall voltage is:

$$V_H = E_H \cdot w = v_d B_0 w$$

Substituting $v_d = \frac{V_0}{2d_A e \rho L}$:

$$V_H = \frac{V_0 B_0 w}{2 d_A e \rho L}$$

Which side is at higher potential? Electrons accumulated on the $+\hat{z}$ face, so:

- The $+\hat{z}$ face is at *lower* potential (negative).
- The $-\hat{z}$ face is at *higher* potential (positive).

Equivalently: $\vec{E}_H$ points in $+\hat{z}$, and electric field always points from high to low potential, confirming that the $-\hat{z}$ side is at higher potential.

Explanation & Reasoning

The Hall voltage is the line integral of the Hall field across the width: $V_H = \int \vec{E}_H \cdot d\vec{\ell} = E_H w$. Combining with the equilibrium condition $E_H = v_d B_0$ from part (d) and the expression for v_d from part (c) yields the final answer.

Sign / side check: The face the electrons *accumulate on* is at the *lower* potential (it has excess negative charge). Always trace which face the carriers move *toward* under the magnetic force.

Grading Rubric

- **+1 pt** – Sets up $V_H = E_H \cdot w$ correctly.
- **+1 pt** – Uses equilibrium $E_H = v_d B_0$ from part (d).
- **+2 pts** – Correct final expression $V_H = V_0 B_0 w / (2 d_A e \rho L)$.
- **+2 pts** – Correctly identifies that the $-\hat{z}$ face is at higher potential (with brief justification).

Problem 6 Solution

a)

$$\frac{Q}{C} - L \frac{dI}{dt} = 0$$

$$I = -\frac{dQ}{dt} \quad \text{since charge is leaving the capacitor}$$

$$\frac{Q}{C} + L \frac{d^2Q}{dt^2} = 0$$

$$\Rightarrow \frac{d^2Q}{dt^2} + \frac{Q}{LC} = 0$$

b)

The second derivative of sine or cosine is minus itself. Alternatively, notice that the differential equation for simple harmonic motion is $\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$, with solution sine or cosine, has the same form as the one above.

c)

$$-\omega^2 + \frac{1}{LC} = 0 \quad \Rightarrow \quad \omega = \frac{1}{\sqrt{LC}}$$

$$Q(0) = Q_0,$$

$$I(0) = 0 \quad \text{since the inductor resists instantaneous change in current}$$

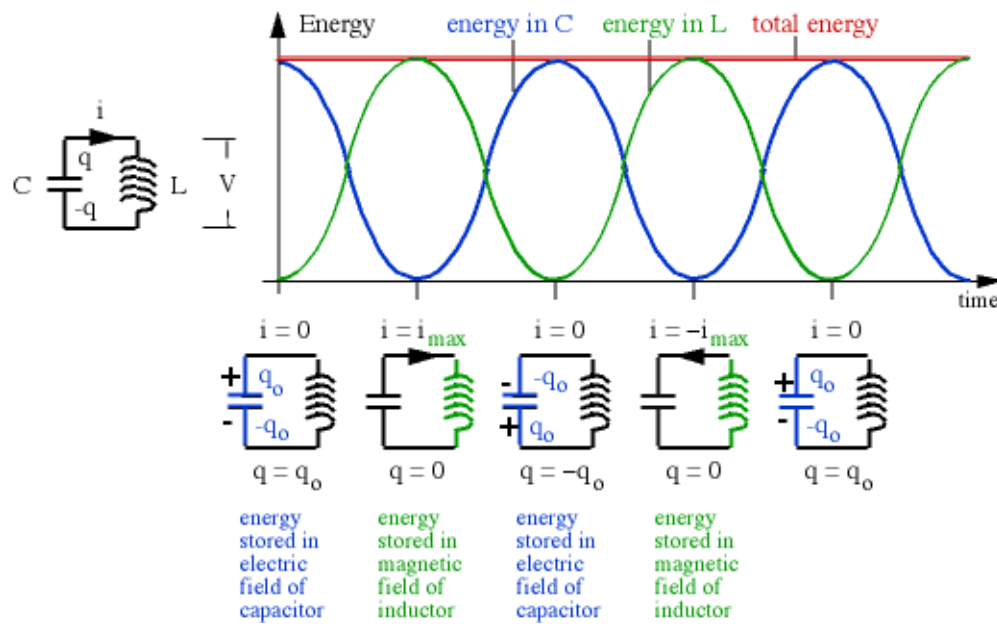
$$\Rightarrow A = Q_0$$

$$\phi = 0$$

d)

$$\begin{aligned} U_E(t) &= \frac{Q(t)^2}{2C} \\ &= \frac{Q_0^2}{2C} \cos^2\left(\frac{t}{\sqrt{LC}}\right) \end{aligned}$$

$$\begin{aligned} U_B(t) &= \frac{1}{2}LI(t)^2 \\ &= \frac{1}{2}L \left[\frac{Q_0}{\sqrt{LC}} \sin\left(\frac{t}{\sqrt{LC}}\right) \right]^2 \\ &= \frac{Q_0^2}{2C} \sin^2\left(\frac{t}{\sqrt{LC}}\right) \end{aligned}$$



e)

No energy is dissipated in the circuit due to the absence of resistance. Analogously, no energy is dissipated in simple harmonic motion with no friction or drag force. If one makes the connection between charge Q and position x , $U_B(t)$ and $U_E(t)$ are analogous to the kinetic and potential energy, respectively, of a simple harmonic oscillator.

Problem 6 Rubric

a.

- 1 pt: Kirchoff's loop law: $+\frac{Q}{C} - L\frac{dI}{dt} = 0$
- 1 pt: $I = -\frac{dQ}{dt}$ for discharging capacitor
- 1 pt: $\frac{d^2Q}{dt^2} + \frac{Q}{LC} = 0$ or something equivalent
- 1 pt: Fully correct/correct Kirchoff's loop law up to a sign

b.

- +1 pt: Identifies second derivative of sine/cosine is proportional to itself
- +1 pt: Identifies taking two derivatives of sine/cosine introduces a negative sign

OR

- +1 pt: Makes connection to simple harmonic motion
- +1 pt: Makes connection between charge Q and position x

c.

- +1 pt: $Q(t=0) = 0$
- +1 pt: $I(t=0) = 0$
- +1 pt: $A = Q_0$
- +1 pt: $\omega = \frac{1}{\sqrt{LC}}$
- +1 pt: $\phi = 0$

d.

- +1 pt: $U_E = \frac{Q^2}{2C}$
- +1 pt: $U_B = \frac{1}{2}LI^2$
- +1 pt: $U_E(t) = \frac{Q_0^2}{2C} \cos^2\left(\frac{1}{\sqrt{LC}}t\right)$
- +1 pt: $U_B(t) = \frac{Q_0^2}{2C} \sin^2\left(\frac{1}{\sqrt{LC}}t\right)$
- +1 pt: Graph shows U_E has a maximum at $t = 0$
- +1 pt: Graph shows U_B has a minimum at $t = 0$
- +1 pt: Graph makes it (approximately) clear U_E and U_B have a phase difference of $\frac{\pi}{2}$
- +1 pt: Graph makes it (approximately) clear U_E and U_B have the same amplitude
- +1 pt: Graph has U_E and $U_B \geq 0$ everywhere

e.

- +1 pt: Mentions no energy dissipated due to $R = 0$
- +1 pt: Mentions simple harmonic motion
- +1 pt: Connection made between U_B and kinetic energy
- +1 pt: Connection made between U_E and potential energy
- +1 pt: Fully correct/gives example of simple harmonic motion without seeing bigger picture

Grading Summary

Part	Points	Key concept
(a) Resistance	4	$R = \rho L/A$, correct cross-section
(b) Current density	4	$j = V_0/(\rho L)$, direction $-\hat{x}$
(c) Drift velocity	5	$n = 2d_A$, electrons drift opposite to $\vec{j}$
(d) Forces & outcome	6	Lorentz force, charge separation, equilibrium
(e) Hall voltage	6	$V_H = E_H w$, identify high-potential face
Total	25	

General Grading Guidelines

- **Carried errors:** If a student makes a mistake in an early part but uses their (incorrect) result *consistently and correctly* in subsequent parts, do not double-penalize. Award full credit for downstream parts if the method is correct.
- **Direction conventions:** If a student consistently uses a different sign convention (e.g. defines current as flowing in $+\hat{x}$), accept the answer as long as the physics is internally consistent.
- **Missing direction labels:** For questions explicitly asking for direction (b, c, d, e), missing direction costs 1 pt.